
JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) characterize input perturbations that change a model prediction, but existing generators often alter more feature coordinates than are required for the desired output transition, thereby reducing interpretability and operational feasibility. We introduce *COLA*, a model- and generator-agnostic post-hoc refinement framework that maps factual and counterfactual sets through an optimal-transport (OT) coupling and uses the resulting joint distribution to parameterize a Shapley attribution rule (p -SHAP) for sparse edit selection. Theoretically, OT minimizes an upper bound on the W_1 divergence between factual and target counterfactual outcomes; under mild regularity conditions, the refined counterfactuals are guaranteed not to be farther from the factual data than the original counterfactual proposals. Empirically, across four datasets, twelve predictive models, and five CE generators, COLA preserves the target effects while using only 26–45% of the original feature edits. On a tractable small-scale benchmark, COLA attains near-optimal performance.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) provides methodological tools for rendering artificial-intelligence systems transparent, auditable, and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods, including Shapley-value approaches (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify the contribution of each input coordinate to a machine learning (ML) model output and thereby identify the variables that dominate a prediction. In healthcare applications, for example, Shapley values can isolate factors such as age and medical history, supporting clinical interpretation of model decisions (Ter-Minassian et al., 2023; Nohara et al., 2022). A complementary class of methods, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), specifies how feature values would need to change to obtain an alternative model outcome. Although hundreds of CE algorithms have been proposed (Guidotti, 2022; Verma et al., 2020), no single generator is suitable for all use cases because objectives, constraints, and model assumptions vary substantially. Some methods construct one counterfactual instance for each factual instance, whereas others operate on groups and return one or more counterfactuals per factual observation. Other formulations act at the dataset level, producing global CE that indicate directions in feature space for achieving a desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). Distributional methods instead treat factual observations as an empirical distribution and search for a counterfactual distribution with comparable geometry and cost structure (You et al., 2025). Algorithmic assumptions also differ: some CE methods require differentiability, whereas others target tree-based or ensemble predictors.

Problem Description and Challenges We study a general refinement problem that builds on the foundations reviewed in Appendix A; the same related-work context is summarized in Appendix A. The central question is how to combine attribution information with counterfactual proposals so that the resulting action plan is both contrastive and sparse. In isolation, FA identifies influential coordinates but does not prescribe concrete edits, whereas CE provides candidate edits that may include unnecessary or infeasible feature changes. We therefore ask:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
$\textcircled{S}$	$\textcircled{C}$	$\textcircled{R}$	$\textcircled{S}$	$\textcircled{C}$	$\textcircled{R}$	$\textcircled{S}$	$\textcircled{C}$	$\textcircled{R}$	$\textcircled{R}$
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] The platform seeks to increase user registrations using observed interaction variables, including spending ($\textcircled{S}$), click count ($\textcircled{C}$), and registration status ($\textcircled{R}$). In the factual matrix ($\mathbf{x}$), no user is registered. The action plans $\mathbf{z}'$ and $\mathbf{z}''$ modify user-level features to approach the target outcome ($\mathbf{y}^*$) of full registration. Both plans attain one-half of the counterfactual effect, but $\mathbf{z}''$ does so with fewer feature modifications than $\mathbf{z}'$.

This problem presents three technical requirements. First, the refinement procedure should not depend on a single CE generator, because counterfactual objectives are task-specific. Second, it should avoid restrictive assumptions on the predictive model, such as differentiability or specialized architecture, so that the procedure remains applicable across model classes. Third, attribution scores used for edit selection must be coherent with the counterfactual objective: generic feature importance can be misleading when it is computed independently of the target counterfactual transition. Equivalently, it is insufficient to perform FA independently of CE to select the most important features to change. We later show in Result II of Section 6 that such decoupling can select counterproductive user or stakeholder actions, because globally important features need not coincide with the coordinates that realize the desired counterfactual outcome.

Main Contributions We propose COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a post-hoc refinement framework that is agnostic to both the predictive model and the CE generator. COLA uses an optimal transport (OT)-induced coupling between factual and counterfactual sets to define Shapley attributions that are explicitly conditioned on factual-counterfactual alignment. The resulting p -SHAP construction subsumes commonly used Shapley variants under appropriate couplings and provides a modular interface for attribution, candidate-value assignment, and sparse edit selection. We establish that OT minimizes an upper bound on the W_1 divergence, and we prove that, under mild conditions, the refined counterfactuals are no farther from the factual data than the original counterfactual proposals. Across four datasets, twelve models, and five CE methods, COLA with p -SHAP preserves the same target effects while using only 26–45% of the original feature edits. On a small-scale benchmark, COLA is near-optimal.

To the best of our knowledge, this is the first systematic method for the problem in Figure 1 that does not require a specified CE algorithm or a specified predictive-model class.

2 PROBLEM FORMULATION

We now formalize the problem represented in Figure 1. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ denote a black-box ML model, and let $\mathbf{x} \in \mathbb{R}^{n \times d}$ denote the factual data matrix with $n \geq 1$ observations and $d \geq 1$ features. Let $\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$, denote the target model output. The optimization variable $\mathbf{z} \in \mathbb{R}^{n \times d}$ represents the refined counterfactual data, and the budget C upper-bounds the number of feature coordinates that may be changed. Given a divergence function D that measures dissimilarity between two objects, the goal is to choose $\mathbf{z}$ so that $f(\mathbf{z})$ is close to $\mathbf{y}^*$ while respecting proximity and sparsity constraints:

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - Mc_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) together with the proximity constraint equation (1b) encodes the standard CE requirement: $\mathbf{z}$ should induce outputs close to $\mathbf{y}^*$ while remaining close to the factual data $\mathbf{x}$. Under this interpretation, $\mathbf{z}$ provides the contrastive reference explaining why $f(\mathbf{x})$ does not attain $\mathbf{y}^*$. We keep D abstract; admissible choices include Euclidean distance, OT, maximum mean discrepancy (MMD), and divergences between mean or median model outcomes. Constraints equation (1c)–equation (1e) impose the action budget. The binary indicator $\mathbf{c}$ determines which entries of $\mathbf{x}$ may be modified: $c_{ik} = 0$ enforces $z_{ik} = x_{ik}$ through equation (1d) and equation (1e), whereas $c_{ik} = 1$ relaxes this equality through the big- M bounds. Constraint equation (1c) permits at most C modified coordinates. The proximity constraint equation (1b) is therefore the formal mechanism that compares $\mathbf{z}$ with $\mathbf{x}$, while the output term compares $f(\mathbf{z})$ with $\mathbf{y}^*$; throughout the subsequent construction, $\mathbf{x}$ is the factual input, $\mathbf{z}$ is the sparse refinement, and $\mathbf{r}$ is the counterfactual proposal used to parameterize candidate edits.

To approximate equation (1), we use FA to estimate which feature coordinates should be activated in the modification indicator $\mathbf{c}$. The next section reviews Shapley-value attribution rules that, together with the proposed rule, are incorporated into the COLA framework for constructing the refined counterfactual $\mathbf{z}$. The optimization problem is computationally difficult even for linear models with $d = 1$; see Appendix B.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews Shapley-value constructions for FA, which serve as baseline attribution modules inside COLA. We first state the cooperative-game definition of the Shapley value and then describe the baseline, random-baseline, and counterfactual variants used for feature attribution. A broader taxonomy of Shapley methods is provided by (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game is specified by a finite player set—here, the feature index set $\mathcal{F} = \{1, 2, \dots, d\}$ —and a characteristic function v . The mapping $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ assigns each coalition $S \subseteq \mathcal{F}$ a payoff $v(S)$, with $v(\emptyset) = 0$. The Shapley value distributes the grand-coalition payoff $v(\mathcal{F})$ across individual players according to their average marginal contributions.

For a player (feature) k and a coalition S , define $\Delta(k, S)$ as the marginal contribution obtained by adding k to S :

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value for feature k is then defined by equation (3).

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

Thus, ϕ_k averages the marginal contribution $\Delta(k, S)$ of feature k over all coalitions that exclude k , with each coalition size normalized by the inverse binomial factor $\binom{d-1}{|S|}$. The resulting quantity is the standard axiomatic allocation of the total payoff to feature k .

For any data point $\mathbf{x}_i$, let $\mathbf{x}_{i,S}$ denote the restriction of $\mathbf{x}_i$ to features in S , and let $\mathbf{x}_{i,\mathcal{F} \setminus S}$ denote the complementary coordinates. The same indexing convention is used for other vectors. We next instantiate v for common Shapley attribution rules.

Baseline Shapley (B-SHAP). This method fixes a baseline point $\mathbf{r}_j$ as the reference for the factual point $\mathbf{x}_i$. The corresponding set function is

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where absent features are imputed using their values in the reference point $\mathbf{r}_j$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This construction assumes an exact alignment between $\mathbf{x}_i$ and $\mathbf{r}_j$.

Random Baseline Shapley (RB-SHAP). RB-SHAP replaces the fixed reference point with samples from a background distribution and is used implicitly in (Lundberg & Lee, 2017) and in the widely used “SHAP” library. Its set function is

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} \left[f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S}) \right] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ denotes the background distribution, typically instantiated by the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). CF-SHAP extends RB-SHAP by replacing $\mathcal{D}$ with a counterfactual distribution conditioned on each factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{\text{CF}}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This construction has been used in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and has demonstrated advantages for contrastive FA. It requires a probabilistic alignment between factual points $\mathbf{x}$ and counterfactual points $\mathbf{r}$.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4) through equation (6) by introducing an alignment module A_{Prob} that returns a joint probability measure over factual and counterfactual instances.¹ The proposed p -SHAP set function is

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

In contrast to baseline methods that rely on random reference draws, p -SHAP explicitly resolves the alignment problem by computing an OT coupling. Feature attribution is therefore formulated not only as a decomposition of predictive output, but also as a transport problem in which the explanatory reference distribution is selected to minimize alignment cost.

The module A_{Prob} is independent of the CE algorithm, but only depends on the generated counterfactual $\mathbf{r}$. It computes $\mathbf{p}$ as the coupling between $\mathbf{x}$ and $\mathbf{r}$, so each entry of $\mathbf{p}$ specifies how mass from a factual row in $\mathbf{x}$ is assigned to candidate rows in $\mathbf{r}$. Conditioning only on the fixed factual and generated counterfactual sets makes p -SHAP invariant to implementation-specific variability in the CE generation process; this is a primary distinction from CF-SHAP. In Result II of Section 6, we show that OT can be more effective than relying on a counterfactual distribution defined by a CE generation mechanism as done by Albini et al. (2022).

A central instantiation of A_{Prob} in this paper is the following OT problem, equivalently the entropically regularized 2-Wasserstein objective. The transport plan $\mathbf{p}_{\text{OT}}$ obtained from this optimization is used as the joint distribution between $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here, $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ are the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$, respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ denotes the feasible set of joint distributions, or transport plans. The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularizer with coefficient $\varepsilon \geq 0$. Choosing $\varepsilon > 0$ yields the regularized OT problem and improves computational efficiency.

Theoretical Aspects of p -SHAP The OT plan minimizes the expected feature-space cost of transporting factual instances $\mathbf{x}$ toward counterfactual instances $\mathbf{r}$ under the selected alignment. This objective is aligned with our refinement goal: identify feature modifications that preserve the counterfactual outcome while minimizing modification cost. Under a Lipschitz-continuity assumption on the predictive model f , this relationship can be stated formally. Theorem 4.1 (proved in Appendix C) shows that the plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant of f multiplied by the square root of the expected transport cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where

¹First, p -SHAP reduces to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment for any pair $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP reduces to RB-SHAP in equation (5) when A_{Prob} is independent of CE and is associated with an arbitrary background distribution $\mathcal{D}$. Third, p -SHAP reduces to CF-SHAP in equation (6) when A_{Prob} is specified by a known CE distribution.

each p_{ij} specifies how factual instance $\mathbf{x}_i$ is aligned with one or more counterfactual instances $\mathbf{r}_j$. Consequently, within p -SHAP, $\mathbf{p}_{OT}$ provides the minimum-cost alignment strategy among feasible plans ($\mathbf{p} \in \Pi$), and this alignment induces the tightest upper bound, within this cost-based relaxation, on the counterfactual-effect discrepancy $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Theorem 4.1 (*p*-SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Equivalently, $\mathbf{p}_{OT}$ minimizes the expected-modification-cost upper bound on $W_1(f(\mathbf{x}), \mathbf{y}^*)$.

Although Theorem 4.1 controls a transport-cost relaxation, the relaxation functions as a convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem. By concentrating attribution mass on low-cost alignment paths, the OT coupling guides the greedy selection step in Algorithm 1 toward excluding non-essential features. Theorem 4.2 states the interventional interpretation of p -SHAP: the set function attributes output changes to interventions on feature subsets (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Moreover, p -SHAP preserves the relevant Shapley axioms satisfied by B-SHAP and RB-SHAP, making it a principled attribution rule. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the standard axiomatic arguments.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch Algorithm 1 specifies COLA as an approximate solver for equation (1). The framework is decomposed into four algorithmic modules. We next describe Algorithm 1 and its module-level data dependencies, with Figure 2 providing a schematic execution trace. At this stage, $\mathbf{x}$ supplies the factual coordinates, $\mathbf{r}$ supplies feasible counterfactual values, and $\mathbf{z}$ records the edited output produced under the sampled action mask; equivalently, $\mathbf{x}$ defines the unchanged baseline, $\mathbf{x}$ determines which coordinates remain fixed, $\mathbf{x}$ anchors the proximity comparison, and $\mathbf{x}$ is modified only through values sourced from $\mathbf{r}$, with $\mathbf{r}$ defining the admissible value source, $\mathbf{r}$ serving as the candidate counterfactual reference, and $\mathbf{x}$ and $\mathbf{r}$ together specifying the refinement domain.

Line 1 (*Applying a CE algorithm to find a counterfactual $\mathbf{r}$*). The generator A_{CE} receives the predictive model f , factual data $\mathbf{x}$, target output $\mathbf{y}^*$, and tolerance ϵ . It returns a counterfactual proposal $\mathbf{r}$ satisfying $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, so that $\mathbf{r}$ is both target-achieving and proximate to $\mathbf{x}$.

Line 2 (*Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). The alignment module A_{Prob} takes $\mathbf{x}$ and $\mathbf{r}$ as input and outputs a nonnegative matrix representing a joint distribution over the n factual rows and m counterfactual rows. This matrix $\mathbf{p}$ encodes the factual-counterfactual matching used in Line 5 to construct candidate replacement values for $\mathbf{z}$. As discussed in Section 3, when no row-level alignment is known, A_{Prob} can be instantiated by OT to obtain the minimum-cost coupling between $\mathbf{x}$ and $\mathbf{r}$. When reliable alignment information is available, A_{Prob} should instead encode that information directly.

Lines 3–4 (*p*-SHAP FA). We apply equation (7) to compute Shapley values for the entries of $\mathbf{x}$. The joint distribution $\mathbf{p}$ may be used to align each factual row with its corresponding counterfactual reference rows, so that the selected rows of $\mathbf{r}$ define the contrastive baseline for the row of $\mathbf{x}$.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

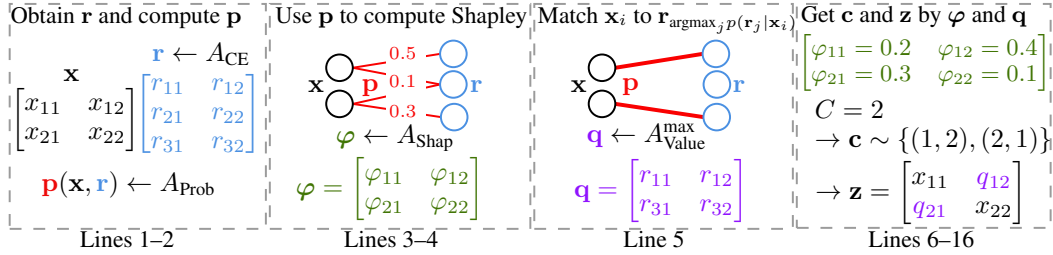


Figure 2: [An illustration of COLA] The figure illustrates how COLA constructs the action indicator $\mathbf{c}$ and the refined counterfactual $\mathbf{z}$ for equation (1). For clarity, line 5 uses $A_{\text{Value}}^{\max}$. In lines 6–16, the budget is $C = 2$, so the sampling step selects exactly two modification coordinates according to the probability matrix φ .

This alignment is empirically important for contrastive FA. The resulting Shapley-value matrix is transformed elementwise by absolute value and normalized to sum to one, producing the attribution probability matrix φ .

Line 5 (Computing feature values). The value-assignment module A_{Value} takes the counterfactual matrix $\mathbf{r}$ and coupling matrix $\mathbf{p}$ as inputs. For each factual row i , it selects one or more rows of $\mathbf{r}$ as aligned references and returns a candidate matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$, where q_{ik} is the proposed counterfactual replacement for x_{ik} ($\forall i, k$). We use $A_{\text{Value}}^{\max}$ for single-reference assignment and $A_{\text{Value}}^{\text{avg}}$ for multi-reference weighted assignment.

For a factual row $\mathbf{x}_i$, $A_{\text{Value}}^{\max}$ selects the counterfactual row with the largest coupling mass.

$$\mathbf{q} = A_{\text{Value}}^{\max}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as the coupling-normalized weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Because φ is nonnegative and normalized to total mass one, it defines a sampling policy over feature coordinates. Line 7 samples C coordinates and sets the corresponding indicators to $c_{ik} = 1$. For each activated coordinate, x_{ik} is replaced by q_{ik} ; the resulting matrix is returned as $\mathbf{z}$ together with the action indicator $\mathbf{c}$, which together constitute the approximate solution to equation (1).

By Theorem 4.1, COLA uses attribution-guided modifications to reduce the discrepancy between $f(\mathbf{z})$ and $\mathbf{y}^*$ while prioritizing coordinates that are most relevant to the desired counterfactual effect. The following theorem (proved in Appendix E) establishes that the refined output $\mathbf{z}$ produced by COLA remains no farther from the factual data than the aligned candidate matrix under the Frobenius norm. Empirical evidence supporting the broader applicability of this proximity behavior is reported in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality states that the refined counterfactual $\mathbf{z}$ is no farther from the factual input $\mathbf{x}$ than the aligned counterfactual proposal $\mathbf{q}$, so the refinement step does not increase Frobenius-distance divergence.

Corollary 5.2. *When $n = m$ and the OT plan $\mathbf{p}$ is a deterministic permutation σ (obtained without entropic regularization, $\varepsilon = 0$), $\mathbf{q}$ is the row-permuted counterfactual matrix $\mathbf{r}_\sigma$, with $q_i = r_{\sigma(i)}$. The proximity bound then reduces to*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the computational complexity of A_{CE} . For A_{Shap} , we consider weighted linear regression for Shapley-value estimation and denote by M_{Shap} the number of sampled subsets. The resulting complexity of COLA, as a function of n, m, d , and the entropic-OT regularization parameter ε , is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ under $A_{\text{Value}}^{\text{max}}$ and $N = \mathcal{O}(nmd)$ under $A_{\text{Value}}^{\text{avg}}$. Appendix F gives the detailed derivation.

6 NUMERICAL RESULTS

This section evaluates COLA as an approximate solution to equation (1). For each experiment, the target is set to $\mathbf{y}^* = f(\mathbf{r})$, where $\mathbf{r}$ is produced by a CE generator A_{CE} . We use four divergence functions: OT measures distribution-level discrepancy, MMD measures discrepancy between distributions through mean embeddings in a high-dimensional feature space, and absolute mean difference (MeanD) and absolute median difference (MedianD) measure differences in mean and median outcomes, respectively. The experiments are designed to evaluate: I) *COLA’s effectiveness for modification minimality*. II) *p-SHAP’s superior performance than other Shapley methods towards modification minimality*. III) *COLA’s near-optimal performance*.

Experiment Setup The benchmark comprises 4 binary-classification datasets, 5 CE algorithms with distinct objectives, and 12 classifiers, as summarized in Table 1. We define an “experiment scenario” as one dataset, one A_{CE} algorithm, and one predictive model f .

The CE generators in Table 1 cover instance-level, group-level, and distributional settings. DiCE and KNN generate counterfactuals for individual factual instances. AReS and GlobeCE operate at the group level and construct a collection of counterfactual instances for the factual data as a whole. Discount treats the factual data as an empirical distribution and estimates a proximate counterfactual distribution.

Table 2 defines the six compared methods. CF- p_{OT} is the proposed p -SHAP configuration, while the remaining methods are baselines. Each method is specified by the choice of A_{Prob} , A_{Value} , and A_{Shap} and is evaluated across the scenarios in Table 1. For example, RB- p_{Uni} sets A_{Prob} to the uniform coupling, uses the resulting $\mathbf{p}$ in equation (9) or equation (10) to compute $\mathbf{q}$, and uses RB-SHAP as A_{Shap} .

The method set is structured as an ablation study. First, RB- p_{Uni} and CF- p_{Uni} differ only in whether counterfactual information is used in attribution. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all use

Table 1: [Experiment Scenarios Setup] The benchmark evaluates five CE algorithms on four datasets and 12 predictive models. A “scenario” is one combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table specifies the 6 compared methods, with colors matched to Figure 3. Each method is evaluated within the scenarios defined in Table 1.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{Prob}: p_{ij} = 1/nm (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{Prob}: \text{Eq. equation (8) (but not used in } A_{Shap})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{Prob}: p_{ij} = 1/nm (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{Prob}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{Prob}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{Prob}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

counterfactual information but differ in the joint distribution supplied to attribution; this isolates the effect of the OT coupling. Third, comparing CF- p_{OT} against RB- p_{OT} tests whether OT improves performance through factual-counterfactual alignment inside A_{Shap} , rather than through the value-assignment stage alone. Finally, CF- p_{Ect} represents the special case in which each counterfactual has a known factual source, yielding an exact alignment and making CF-SHAP coincide with B-SHAP.

Table 3: [COLA for Modification Minimality] The table reports the number of modified features in $\mathbf{z}$ for each method when $f(\mathbf{z})$ attains 80% of the counterfactual effect of $f(\mathbf{r})$. Each entry averages results over 4 randomly selected scenarios from Table 1, with 100 runs per scenario. The symbol “-” indicates that the target counterfactual effect is not achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

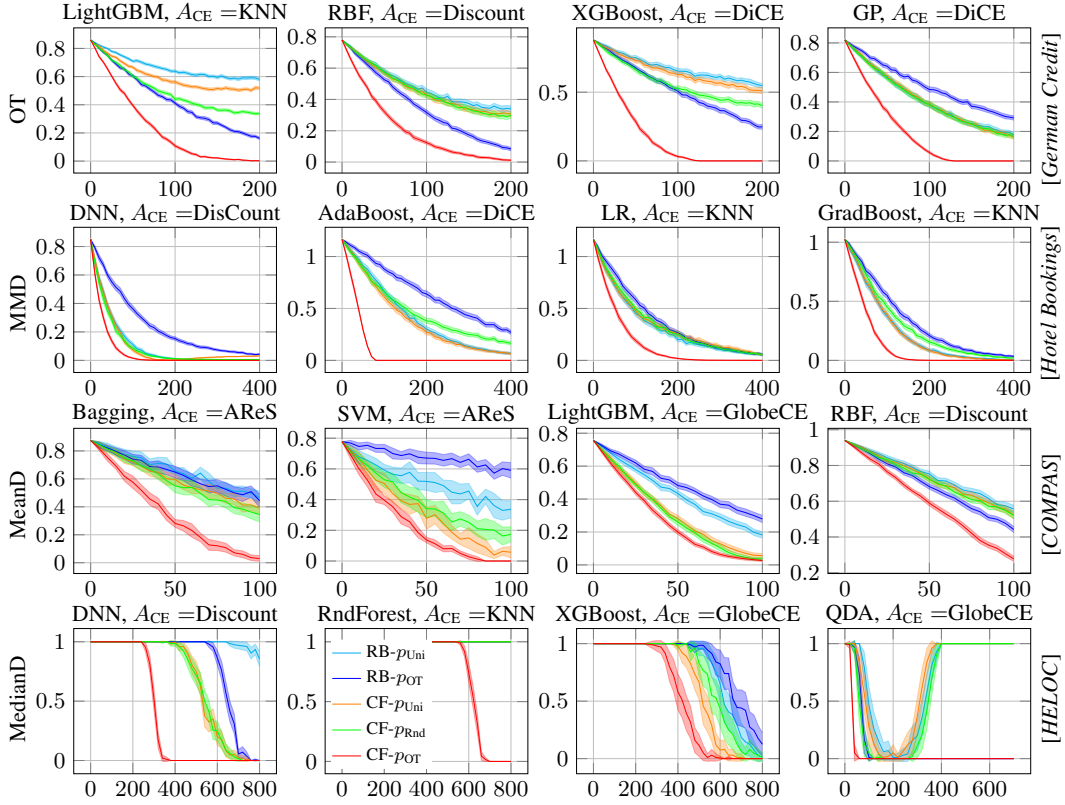


Figure 3: $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the allowed-action budget C . Each curve aggregates 100 runs, and shaded regions denote 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, while $A_{\text{Value}}^{\text{max}}$ is used for *German Credit* and *Hotel Bookings*. The legend in “RnDForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all panels.

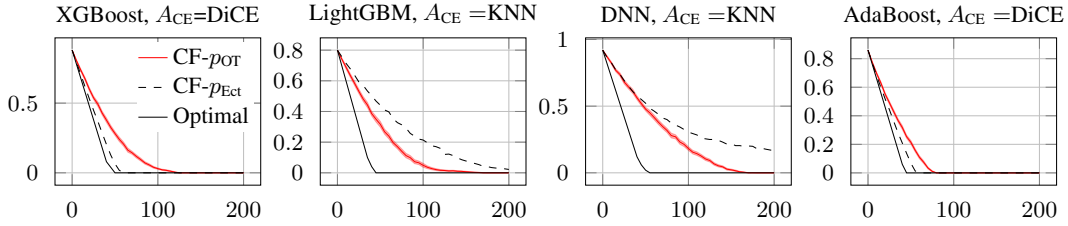


Figure 4: [*German Credit*] $D(f(\mathbf{z}), \mathbf{y}^*)$ as a function of the allowed-action budget C , with D instantiated as *MeanD*.

Result I: COLA achieves significant action reduction with a minor loss in counterfactual effect

In Table 3, the target for $\mathbf{z}$ is defined so that $f(\mathbf{z})$ attains either 80% or 100% of the counterfactual effect of $f(\mathbf{r})$ ². The results show that COLA reaches these targets with substantially fewer actions than the original counterfactual proposal $\mathbf{r}$. For the 80% counterfactual-effect target, COLA uses only 13%–25% of the original feature-change magnitude, measured by $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$. For the 100% target, only COLA with p -SHAP (CF- p_{OT}) succeeds, requiring only 26%–45% of the feature changes in the original $\mathbf{r}$. Thus, p -SHAP yields the strongest action-minimality performance, which we analyze next.

²Note that by definition, we have $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which represents a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. To define a counterfactual effect 80%, consider the proportion of divergence reduced by the refined CE $\mathbf{z}$. That is, $\text{Counterfactual Effect} = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D instantiated as OT.

Result II: p -SHAP outperforms the other Shapley methods in achieving counterfactual effect Figure 3 compares the Shapley variants in equation (4)–equation (7); the horizontal axis is the action budget C , and the vertical axis is $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1). First, $\text{RB-}p_{\text{Uni}}$ and $\text{RB-}p_{\text{OT}}$ perform substantially worse than the counterfactual-informed variants, indicating that CE information is necessary for attribution-based modification minimality. Second, $\text{CF-}p_{\text{OT}}$ improves over $\text{RB-}p_{\text{OT}}$, which isolates the benefit of using OT alignment inside A_{Shap} because the two methods differ only in the Shapley module (see Table 2). Third, p -SHAP outperforms $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$, showing that counterfactual information alone is insufficient: the factual and counterfactual sets must also be coupled through an informative alignment. As Figure 4 later indicates, this alignment need not coincide with the exact alignment induced by the counterfactual generator.

Across the scenarios in Table 1, COLA with p -SHAP consistently exhibits strong performance in both Figure 3 and Table 3. Additional scenario-level results are reported in Appendix H. Overall, p -SHAP provides the best sparsity-oriented refinement among the evaluated Shapley methods.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks This experiment evaluates whether p -SHAP can reduce dependence on CE-generator-specific alignment knowledge by replacing that knowledge³ with the OT joint distribution between factual and counterfactual data, as shown in Figure 4. We benchmark $\text{CF-}p_{\text{Ect}}$ within COLA and solve equation (1) with MeanD as the divergence function D . Because $\text{CF-}p_{\text{Ect}}$ uses a known factual-counterfactual alignment, this comparison tests how effectively COLA exploits exact alignment information. The corresponding optimum can be obtained by solving an MILP; Appendix G derives this formulation. Since the MILP is computationally expensive, we run this benchmark only on *German Credit*.

$\text{CF-}p_{\text{Ect}}$ achieves near-optimal performance with DiCE. For DiCE and KNN, factual-counterfactual pairs are instance-wise independent, indicating that COLA is effective for instance-based CE even though equation (1) is formulated for the more general group or distributional setting. $\text{CF-}p_{\text{OT}}$ sometimes outperforms $\text{CF-}p_{\text{Ect}}$, because it utilizes a more theoretically grounded approach to identify the key features that require modification, whereas $\text{CF-}p_{\text{Ect}}$ relies on CE algorithm-dependent knowledge, which lacks solid justification on its effectiveness for FA. A gap remains between $\text{CF-}p_{\text{Ect}}$, $\text{CF-}p_{\text{OT}}$, and the KNN optimum, indicating that optimal factual-counterfactual alignment remains an open problem.

7 CONCLUSIONS

This paper introduced COLA, a post-hoc framework for refining CE through joint-distribution-informed Shapley values. The framework uses factual-counterfactual couplings to select sparse feature modifications, thereby preserving the counterfactual effect while reducing the number of required actions.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ The paper develops a methodological and empirical refinement procedure (*COLA* with p -SHAP) for CE. The study does not involve human-subject research, interventions, new data collection, annotation of personal records, or release of personal information.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We use these datasets solely for benchmarking and follow the applicable licenses, terms, and common research practices. We do not attempt re-identification or record linkage, and no additional personal information is released. The implementation will include mechanisms for excluding protected attributes and enforcing feasibility or immutability constraints on features.

³This knowledge, i.e. exact alignment between factual and counterfactual, is available only by DiCE and KNN among all CE methods considered.

⁴<https://iclr.cc/public/CodeOfEthics>

CE can reveal biases in underlying models, but they do not by themselves establish fairness. The proposed method reduces the number of recommended feature edits; it does *not* authorize changes to sensitive or immutable attributes (e.g., sex, race, age) and does not provide causal validity. We discourage using CE to manipulate high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals toward infeasible or unethical behavioral changes. Deployment in consequential domains should include domain-expert feasibility constraints, fairness audits (e.g., disparate impact or benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. The attribution produced by the method is distributional rather than causal and must not be interpreted as a causal effect.

The primary risks are (i) interpreting distributional attributions as causal claims, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation of decision systems. We mitigate these risks by (a) explicitly stating the non-causal scope and limitations, (b) supporting cost and feasibility masks together with immutable-feature constraints, (c) reporting subgroup performance where relevant, and (d) documenting responsible-use guidance.

To support replication, we provide an anonymized implementation, detailed experimental settings, and random seeds. We report model classes, hyperparameters, compute budgets, and fixed evaluation protocols to reduce the risk of selective reporting. To our knowledge, this submission has no conflicts of interest or sensitive sponsorships; any funding disclosures will be provided after double-blind review.

REPRODUCIBILITY STATEMENT

We specify the components required for reproducibility. The COLA framework and p-SHAP attribution rule are defined through notation and pseudocode in the main paper (Algorithm 1; Method and Preliminaries sections), while theoretical assumptions and full proofs are provided in the appendix.

Appendix I documents hyperparameters, optimization settings, software-environment details, random seeds, datasets, preprocessing, and data splits. The experimental sections and ablations enumerate baselines, CE generators, predictive models, and coupling choices, with matching configurations in the supplemental materials. The experiment code and software will be released upon acceptance. Hardware, runtime, ablation and sensitivity studies, and the small-scale MILP benchmark specification are also reported in Appendix I to support independent verification.

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A COMPARISON TO EXISTING APPROACHES

The authors in (Albini et al., 2022) proposed CF-SHAP, which uses counterfactual data points as the Shapley background distribution, but it assumes that this counterfactual distribution is conditioned on each individual factual instance. This assumption implies a known probabilistic alignment between factual and counterfactual instances; related assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). Such an alignment is generally unavailable in global, group-level, or distributional explanation settings. The construction in (Albini et al., 2022) is therefore a special case of our framework, and the conditional-counterfactual assumption in (Albini et al., 2022) is not required by the proposed coupling-based formulation. WeightedSHAP (Kwon & Zou, 2022) modifies feature-level contributions by assigning feature weights. Our method instead weights data-point contributions through a joint distribution and can be extended to weight both rows and columns. The framework in (Kommiya Mothilal et al., 2021) combines FA and CE, but its CE-based attribution inherits the same conditional-counterfactual assumption, limiting direct extension to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). Another related method (Sharma et al.) uses a system causal structure and time-series predictors to decompose a single observed change in a system-level metric into input-level contributions. These lines of work do not address the minimal-action CE problem studied here.

The formulations in (Kanamori et al., 2022; Karimi et al., 2021) are closely related to our setting. (Karimi et al., 2021) minimizes action cost under the assumption that a structural causal model (SCM) is known, an assumption that is rarely satisfied in practice. (Kanamori et al., 2022) notes that whether existing CE methods can solve this class of problems remains open, and proposes an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Because MILP solving is computationally expensive, such exact approaches are difficult to scale.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 shows that a scalable exact algorithm for the main optimization problem should not be expected in general unless $\mathcal{P} = \mathcal{NP}$. The hardness is not solely due to nonlinearity of f ; it also follows from the combinatorial sparsity constraint.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem. Given $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and sparsity level $K \in \mathbb{N}$, the task is to find $\mathbf{z} \in \mathbb{R}^n$ with at most K nonzero entries while minimizing squared residual error:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ counts the nonzero entries of $\mathbf{z}$. The problem is $\mathcal{NP}$ -hard because solving it requires selecting a cardinality-constrained subset of active variables.

We reduce this SR problem to problem equation (1) by the following construction. Set the number of features to $d = 1$ and the number of instances to n , matching the dimension of the SR variable. Let the factual data be the zero vector $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$, let the target output be the same $\mathbf{y}^* \in \mathbb{R}^m$ as in the SR instance, and define the predictive model by the linear map $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

Define the output divergence as squared Euclidean loss, $D(f(\mathbf{z}), \mathbf{y}^*) = \|f(\mathbf{z}) - \mathbf{y}^*\|_2^2$. Define the instance-space divergence as $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, using $\mathbf{x} = \mathbf{0}$. Set the action budget C equal to k , the sparsity level in the SR instance, and choose M as any sufficiently large positive constant, for example $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

Thus, $c_i = 0$ imposes both $z_i \leq 0$ and $z_i \geq 0$, forcing $z_i = 0$. When $c_i = 1$, the variable is allowed to range over $[-M, M]$. Constraint equation (1c) reduces to $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \quad (11a)$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \quad (11b)$$

$$\sum_{k=1}^d c_i \leq k, \quad (11c)$$

$$z_i = 0, \quad \text{if } c_i = 0, \quad (11d)$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \quad (11e)$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \quad (11f)$$

In this formulation, c_i indicates whether z_i is active. The constraints enforce $z_i = 0$ whenever $c_i = 0$, exactly reproducing the sparsity mechanism in the SR problem. The budget $\sum_{k=1}^n c_i \leq k$ permits at most k active variables, and the objective is identical to the SR objective.

Because the constructed instance of problem equation (1) is equivalent to cardinality-constrained SR, solving problem equation (1) would solve an $\mathcal{NP}$ -hard problem. Hence problem equation (1) is $\mathcal{NP}$ -hard even for $d = 1$, linear f , and Euclidean divergence functions D . $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below provides that feature attributions are aligned with the expected costs of feature modifications, leading to action plans that are cost-efficient in achieving counterfactual outcomes.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs*

between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted as $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Therefore, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\|\mathbf{q} - \mathbf{x}\|_F^2 = \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2,$$

$$\|\mathbf{z} - \mathbf{x}\|_F^2 = \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2.$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{Value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We perform analysis of the computational complexity of COLA as follows.

First, we analyze A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} just constructs the matrix $\mathbf{p}$ with the prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise, we consider solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing

the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\max}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\max}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\max}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section defines a global optimality benchmark for the case in which the factual-counterfactual alignment is known and equation (1) is solved with D instantiated as MeanD:

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Because COLA starts from $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$ and constructs $\mathbf{z}$ no farther from $\mathbf{x}$ than $\mathbf{r}$, constraint equation (1b) can be omitted in this benchmark. Here $\mathbf{x}$ and $\mathbf{r}$ define the known alignment, while $\mathbf{z}$ is optimized as the sparse refinement and $\mathbf{z}$ determines the benchmark objective. The resulting formulation is

$$\min_{\mathbf{c}, \mathbf{z}} \quad \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

The original constraints equation (1d) and equation (1e) collapse into equation (13c) because CF- p_{Ect} imposes exact alignment. For each $\mathbf{x}_i$, an aligned reference $\mathbf{r}_j$ is available to both A_{Shap} and A_{Value} , so each coordinate x_{ik} either remains unchanged or is replaced by r_{jk} . Thus $z_{ik} = r_{jk}c_{ik} + x_{ik}(1 - c_{ik})$, with the binary variable c_{ik} selecting between factual and counterfactual values.

Because the alignment between each $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$ is known, $\mathbf{q}$ is fixed; in this case, $A_{\text{Values}}^{\text{max}}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$. For data point i and feature subset $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denote the candidate obtained by replacing all coordinates $k \in S$ with q_{ik} while keeping all coordinates $h \in \mathcal{F} \setminus S$ equal to x_{ih} . The collection $\{\mathbf{z}_{iS} : S \subseteq \mathcal{F}\}$ enumerates the possible values of $\mathbf{z}_i$. Define the corresponding scalar

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

For any feasible $\mathbf{z}_i$ in equation (13), the aggregate deviation $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can therefore be encoded using binary selection variables a_{iS} and coefficients g_{iS} :

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

This yields the following mixed-integer reformulation of equation (13).

$$\min_{\mathbf{a}, \eta} \quad \eta, \tag{14a}$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \tag{14b}$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \tag{14c}$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \tag{14d}$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \tag{14e}$$

Minimizing η subject to equation (14b) and equation (14c) is equivalent to minimizing the absolute-value objective in equation (13). Constraint equation (14d) enforces that each data point i selects exactly one modification plan a_{iS} , and equation (14e) enforces the same action budget as equation (13b). Solving equation (14) therefore gives the known-alignment optimum used for the comparison in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

Observing Figures 5–8, CF- p_{Uni} generally performs better than RB- p_{Uni} . Second, consider RB- p_{OT} and CF- p_{OT} that also differ only in A_{Shap} , the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

We analyze how different Shapley methods affect FA, corresponding to lines 3–4 in COLA. The shapley methods can be classified into two categories: First, consider RB- p_{Uni} and CF- p_{Uni} that differ only in A_{Shap} . Observing Figures 5–8, CF- p_{Uni} generally performs better than RB- p_{Uni} . Second, consider RB- p_{OT} and CF- p_{OT} that also differ only in A_{Shap} , the latter consistently outperforms the former. Hence *RB-SHAP is not suitable for FA in CE*.

In addition to FA, line 5 is a critical component of COLA: it uses the joint probability $p(\mathbf{x}, \mathbf{r})$ to construct the candidate matrix $\mathbf{q}$, which specifies the counterfactual replacement direction for each factual feature value. In Figures 5–8, CF- p_{OT} consistently outperforms the other methods. The three methods CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all provide candidate joint probabilities $\mathbf{p}$ when exact factual-counterfactual alignment is unavailable, but their performance differs substantially. Thus, knowing only the CE set and its marginal distribution is insufficient.

OT is particularly useful when alignment information between factual and counterfactual instances is missing or inaccurate. Even when a CE algorithm explicitly pairs each factual instance with

a counterfactual counterpart, the induced alignment need not optimize downstream refinement performance. Figure 4 in Section 6 supports this observation.

The coupling p_{OT} need not equal the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$. Its role is operational: it supplies COLA with an alignment that jointly informs FA and CE refinement. The QDA column in Figure 5 further illustrates the stability of OT-based methods, whereas alternative alignments diverge substantially from the target. Nevertheless, COLA is *not limited to using OT* as A_{Prob} . As indicated by Figure 4, even a known best $\mathbf{p}$ can retain a non-negligible gap to the global optimum. Designing stronger A_{Prob} algorithms therefore remains an important direction for future work.

I EXPERIMENTS REPRODUCIBILITY

The experiments were executed on a high-performance computing (HPC) cluster using four parallel nodes, one per dataset. Each node contains two Intel Xeon Processor 2660v3 CPUs (10 cores, 2.60GHz) and 128 GB of memory. Runtime is approximately 5–10 hours per node, depending on dataset size. The experiments can also be reproduced on a laptop, but with higher wall-clock time than on the HPC cluster.

For all four datasets, numerical features are standardized, and categorical features are encoded using either label encoding or one-hot encoding. We did not observe material differences in COLA’s performance across these categorical-encoding choices. The train-test split ratio is 7 : 3.

The optimality baseline in Figure 4 is computed with Gurobi 11.0.2 ([gur](#)). Reproducing this baseline requires a Gurobi license; alternatively, open-source operations-research libraries such as Google-OR Tools ([goo](#)) may be used. The MILP in Appendix G is computationally expensive and is therefore practical only for small-scale datasets such as German Credit. Computing the same baseline on other datasets requires reducing the number of included features.

Model hyperparameters are specified as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are implemented using scikit-learn ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers, and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are accessed through their scikit-learn interfaces with default hyperparameters.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models (LLMs) were used only as general-purpose writing-assistance tools. Their use was restricted to grammar correction, wording refinement, and light copy-editing of author-written text. LLMs did not contribute to research ideation, modeling choices, experimental design, algorithms, proofs, datasets, analyses, or results. All LLM-suggested phrasing was reviewed and edited by the authors before inclusion, and the authors retain full responsibility for the manuscript content. The authors take full responsibility for all content; LLMs are not authors or contributors. This disclosure complies with the ICLR policy on LLM usage.

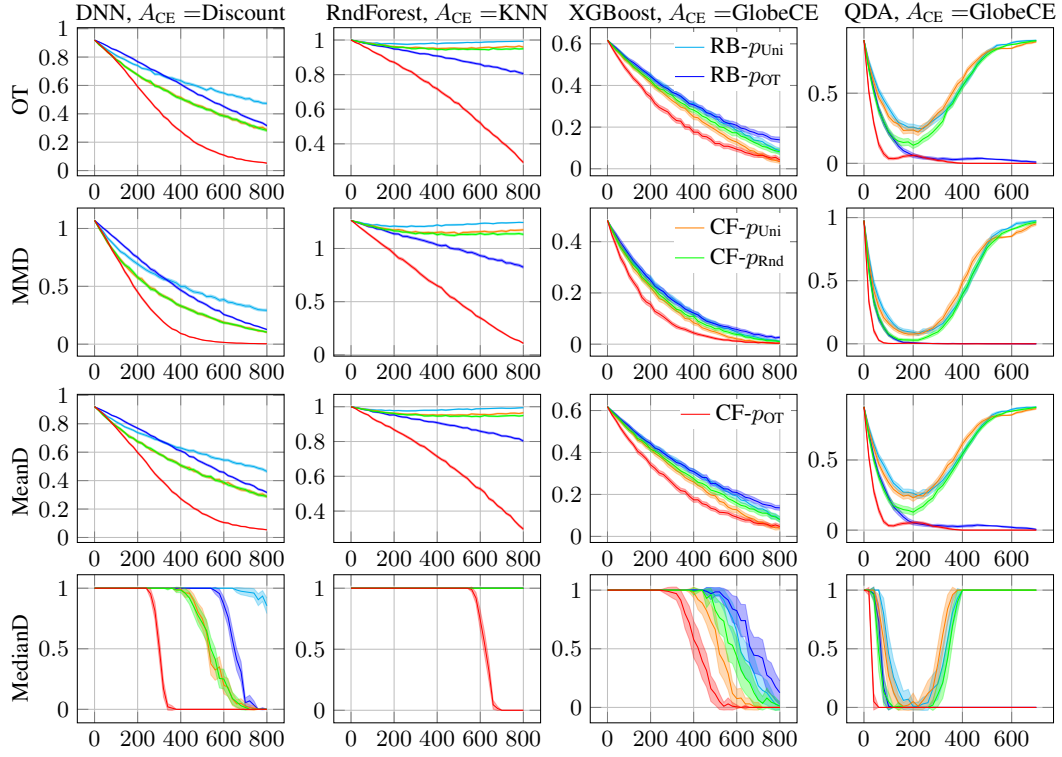


Figure 5: [HELOC] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

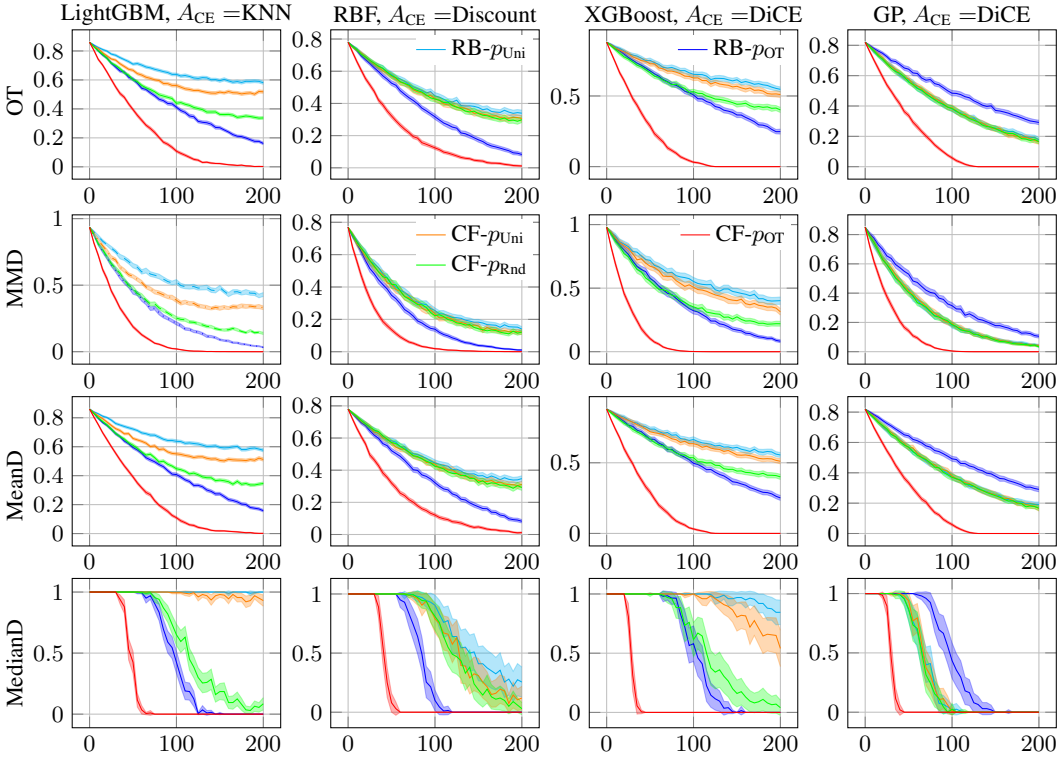


Figure 6: [German Credit] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

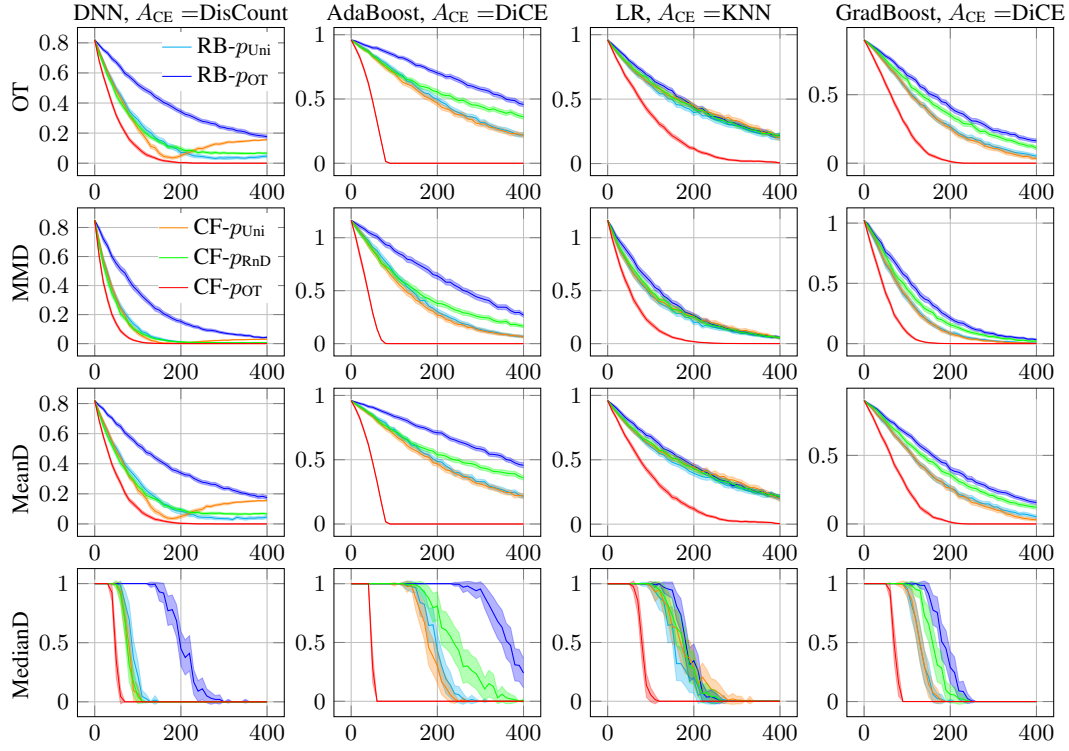


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

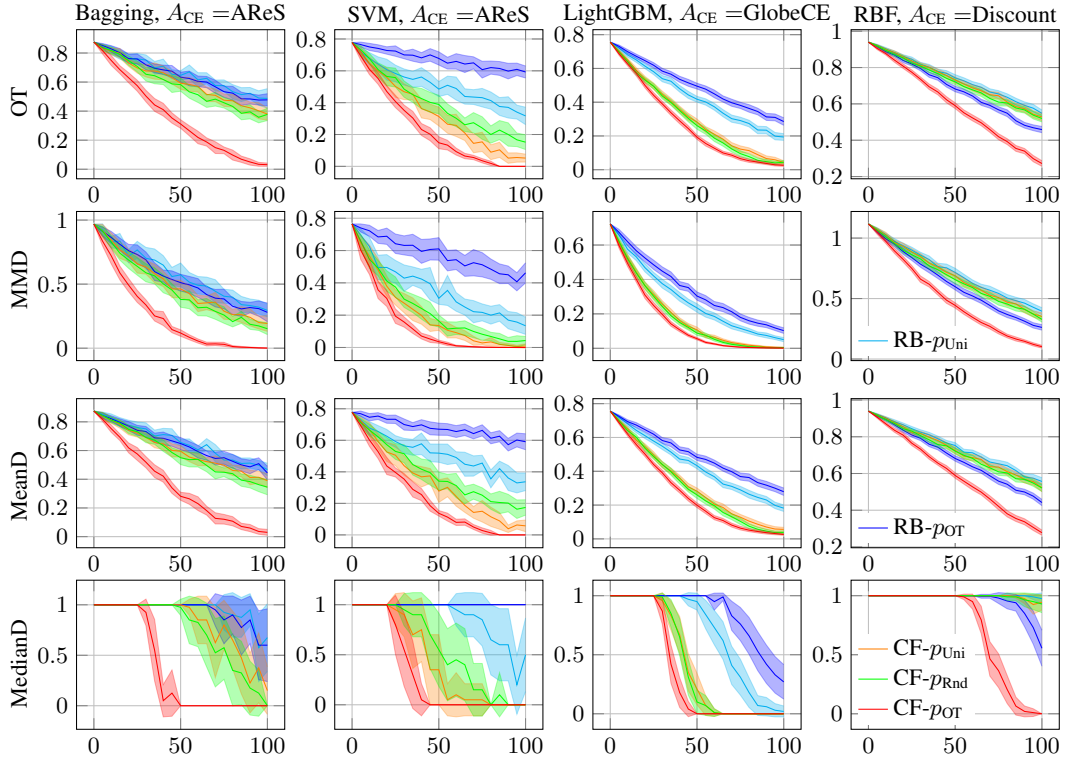


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C . Experiments are with 100 runs. The shadows show the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.